Ex no: 1
11/2/2026

Verification of kirchoff's laws

AIM :-

To verify kirchoff's law of voltage and kirchoff's current law both theoretically and practically for a given DC circuit.

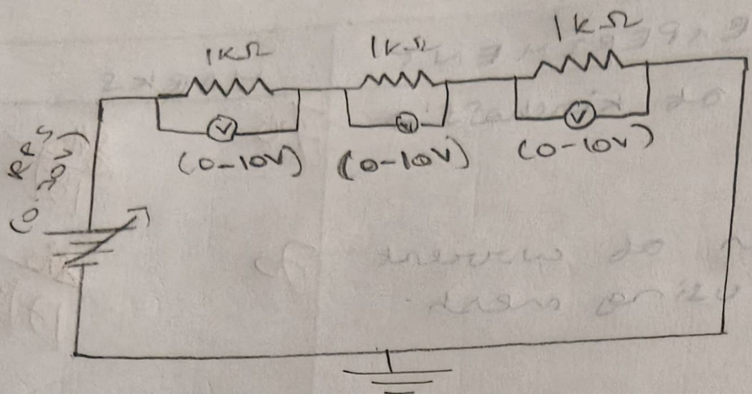
APPARTUS REQUIRED :-

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1)	Regulated Power Supply	(0-30V)	1
2)	Voltmeter	(0-30)V MC	3
3)	Ammeter	(0-10mA) MC	3
4)	Resistors	1kΩ	3
5)	Bread board	-	1

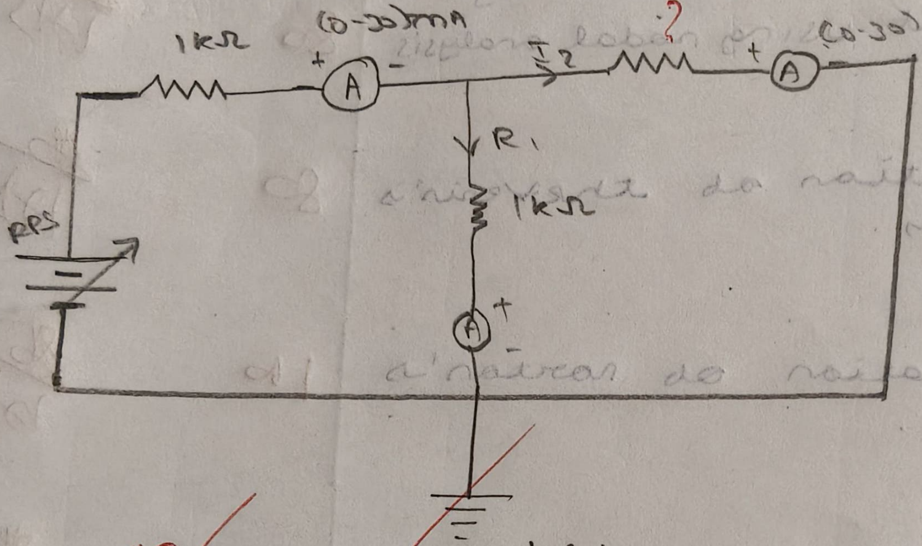
PROCEDURE :-

1) Give connection as per the circuit diagram
2) switch on the supply vary the RPS and set a particular input voltage
3) Note down the readings of ammeter and voltmeter and tabulate them.
4) vary the RPS to its minimum value and switch off the supply
5) using the tabulated values, verify the

Circuit diagram :-



KVL



KCL

V	V_1	V_2	V_3	$V=V_1+V_2+V_3$	I_1	I_2	I_2	$I=I_1+I_2$
2	0.61	0.61	0.61	1.83	0.57	0.58	0.57	1.15
6	1.97	1.97	1.97	5.91	1.67	1.68	1.67	3.35
9	2.89	2.89	2.89	8.68	2.49	2.52	2.49	5.01

O/p verified

Marks - split - UP :-

S.No	Description	Marks allotted	Marks obtained
1)	Experimental set-up correction	20 marks	20
2)	observation	5 marks	20
3)	calculation	5 marks	20
4)	viva	5 marks	20
5)	Result	5 marks	10
		TOTAL -	70

70

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RESULT :-

The kirchoff's current law and voltage law are verified practically and theortically.